

Cricket Player Performance Tracking Application

CRICKSTATS

A PROJECT REPORT

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BACHELOR OF COMPUTER APPLICATION

UNDER THE GUIDANCE OF

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1. INTRODUCTION

1.1 Project Overview :

CRICKSTATS is a Salesforce-based application designed to track, manage, and analyze cricket players' performance on a match-by-match basis. The system uses custom objects, relationships, dashboards, and automation tools to provide structured performance insights.

It enables coaches, analysts, and administrators to record player statistics, generate reports, and support strategic decision-making.

1.2 Purpose :

The main purpose of this project is:

To digitize player performance records

To improve accuracy and accessibility of cricket statistics

To provide real-time analytics

To support player development and team planning

To automate data management using Salesforce

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
PS-2	A team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	A sports analyst	Analyze player trends and generate reports	I face difficulty in comparing performances	There is no centralized analytics system	Confused and dissatisfied

2.2 Empathy Map Canvas

Target Users

- Team Managers

- Coaches
- Analysts
- Salesforce Admins

Empathy Analysis

Aspect	Description
● Think & Feel:	Need accurate data, better insights
● Hear:	Feedback from management
● See:	Multiple data sources
● Say & Do:	Collect statistics manually
● Pain:	Data errors, delays
● Gain:	Faster decisions, automation

2.3 Brainstorming

Brainstormed Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Enter Player Data
- Record Match Performance
- Generate Reports
- Analyze Results
- Decision Making

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements (NFRs) – CRICKSTATS

NFR-1: Usability

The system should be easy to understand and use by coaches, managers, analysts, and admins without requiring technical training.

Requirements

- Simple and intuitive user interface
- Clear navigation menus

Examples:

- A new coach should be able to:
- Log in

NFR-2: Security

The system must protect sensitive player data from unauthorized access.

Requirements

- Role-based access control
- Regular backups

.

Examples:

- Coaches can view player statistics
- Managers can edit records

NFR-3: Reliability

The system should function correctly without failure and store data accurately.

Requirements

- No data loss
- Automatic backups

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

Requirements

- Fast page loading
- Quick report generation

NFR-5: Availability

The system should be accessible whenever users need it.

Requirements

- 24/7 system availability
- Minimal downtime

NFR-6: Scalability

Requirements

- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

3.3 Data Flow Diagram

Data Flow

User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data

were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by using secure cloud storage that safely stores all information and allows easy retrieval.

4.1 Proposed Solution

Description

CRICKSTATS provides:

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries

Key Features

- Centralized database
- Live statistics
- Easy reporting

4.3 Solution Architecture

Architecture Layers

- Presentation Layer (Lightning UI)
- Application Layer (Flows, Logic)
- Data Layer (Objects)
- Analytics Layer (Reports)

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile-based execution
- Object creation → automation → reporting

6. Project Development Phase:

Project Flow

Milestone 1 : Creation of developer account

Milestone 2 : Object Creation

Milestone 3 : Tabs

Milestone 4 : The Lightning App

Milestone 5 : Fields & Relationships

Milestone 6 : Roll-up Summary

Milestone 7 : Flows

Implementation

Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to <https://developer.salesforce.com/signup> 2.

On the sign up form, enter the following details :

- 1) First name & Last name
- 2) Email
- 3) Role : Developer
- 4) Company : College or Company Name
- 5) County : India
- 6) Postal Code : pin code
- 7) Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in that format :

username@organization.com

Click on sign me up after filling these.

Activity 2: Account Activation :

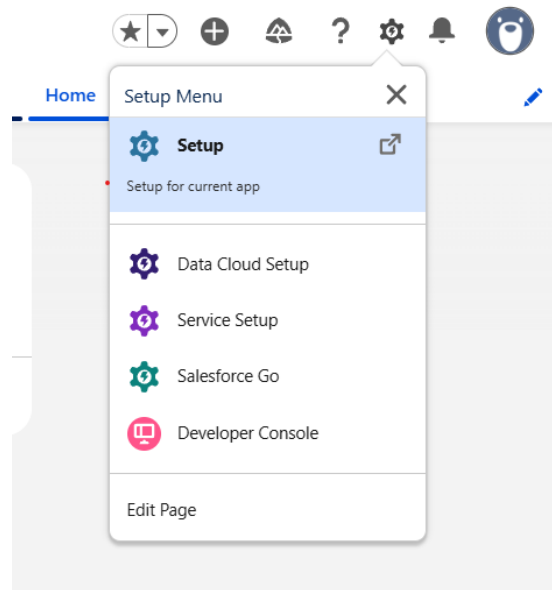
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.
2. Click on Verify Account.

3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.

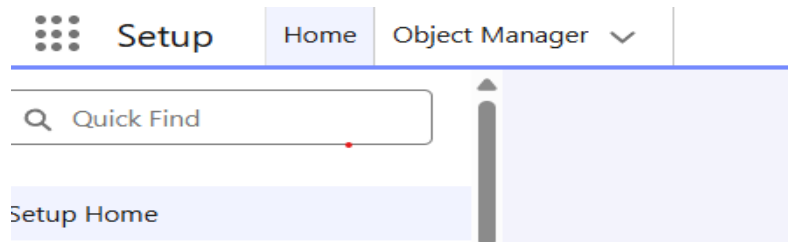
Milestone 2 - Object Creation

Activity 1 : Player Object

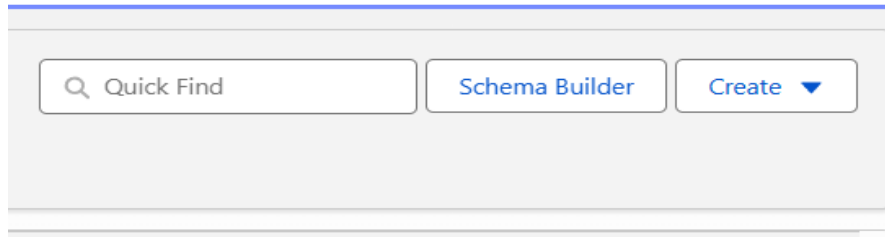
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.



2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label** : Player
- **Plural Label** : Players
- Enter Record Name Label and Format
Record Name : Player Name
Data Type : Text
- **In Optional features** : check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status** : check the box for Allow Search.
- Save

Activity 2 : Match Performance Object

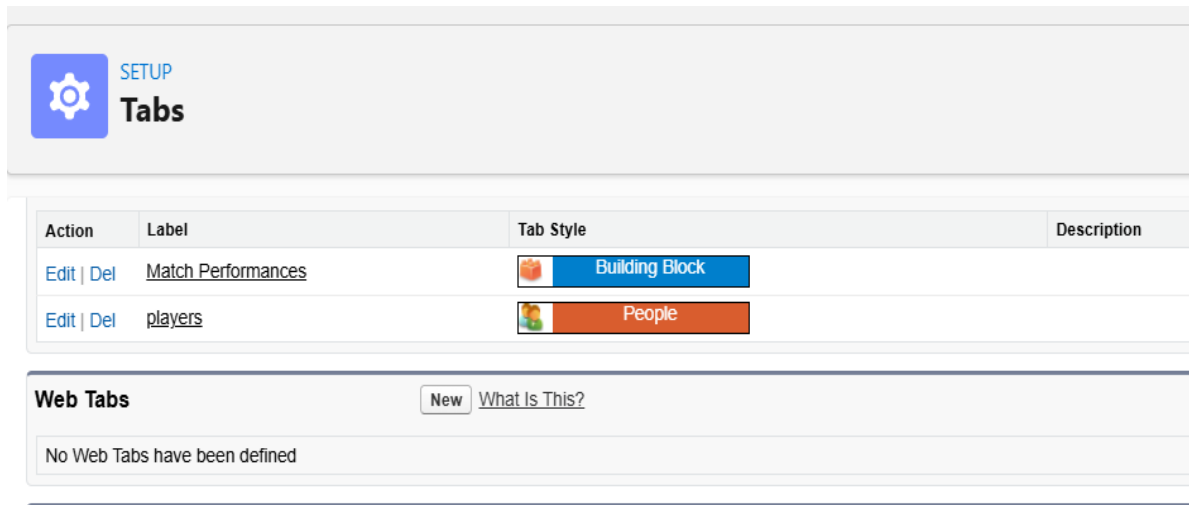
1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label** : Match Performance
- **Plural Label** : Match Performances
- Enter Record Name Label and Format
Record Name : Customer Account Name
Data Type : Text
- **In Optional features** : check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status** : check the box for Allow Search. • Save.

Milestone 3 - Tabs

Activity 1 : Custom tabs creation

1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)
2. Select Object(Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Match Performances.



Action	Label	Tab Style	Description
Edit Del	Match Performances	 Building Block	
Edit Del	players	 People	

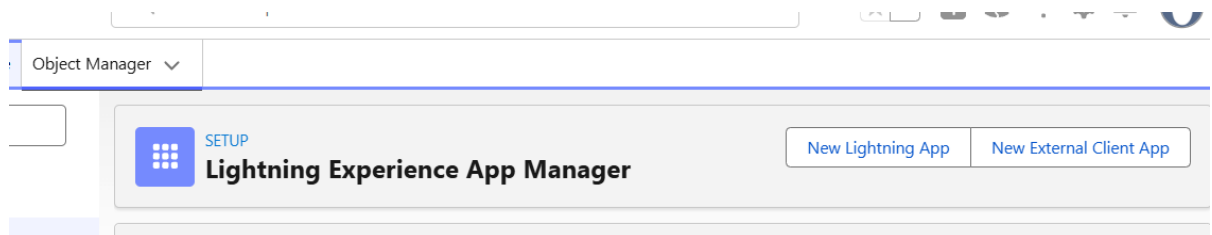
Web Tabs [New](#) [What Is This?](#)

No Web Tabs have been defined

Milestone 4 - The Lightning App

Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.



Object Manager ▾

SETUP Lightning Experience App Manager

[New Lightning App](#) [New External Client App](#)

2. Fill the app name in app details and branding as follow
 App Name : CRICKSTATS
 Developer Name : Auto Populates

Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default. Then click Next.

3. In App options

Navigation style : Standard navigation

Setup experience : setup

Supporters form factors : Desktop and phone

Then click Next

4. In Utility items

Utility Bar alignment : Default Then
click Next.

5. Navigation items

Select the created Custom Objects and required standard objects

- Players
 - Match Performances
 - Reports
 - Dashboards

6. To Add User Profiles : System Administrator

New Lightning App

User Profiles

Choose the user profiles that can access this app.

Available Profiles

- Analytics Cloud Integration User
- Analytics Cloud Security User

Selected Profiles

System Administrator

✓

✓

✓

✓

○

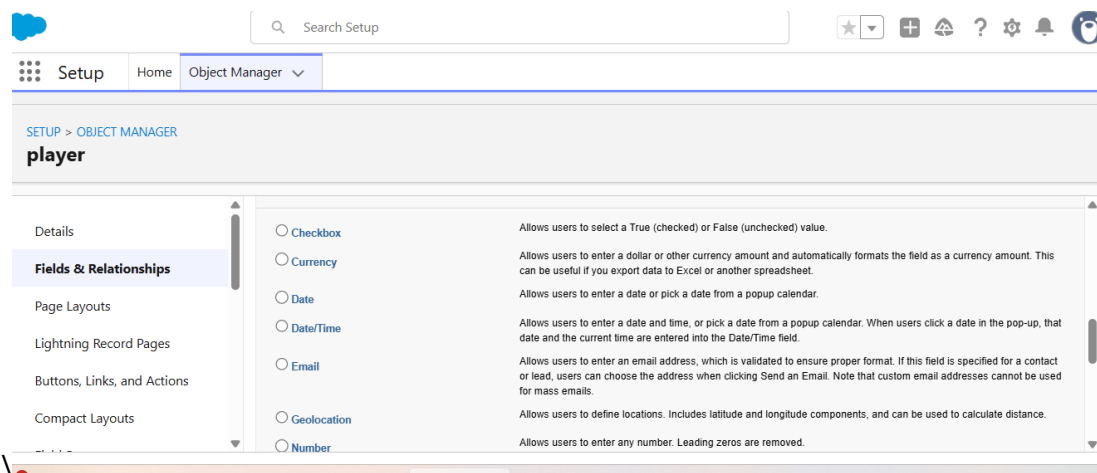
Save & Finish

7. Click Save & Finish.

Milestone 5 : Fields & Relationships

Activity 1 : Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input.



7. Click on next.
8. Data Type : **Text**
Field Label: Player Name
Field Name : Auto Populates
Length : 50
Click on Next → Next → Save and new.

9. Repeat the Same steps for remaining fields

- Data Type : **Text**
Field Label: Player Number
Field Name : Auto Populates
Length : 3
Click on Next → Next → Save and new.
- Data Type : **Rich Text Area**
Field Label : Profile Picture
Field Name : Auto Populates
Click on Next → Next → Save and new.
- Data Type : **Date**
Field Label : Date of Birth
Field Name : Auto Populates
Click on Next → Next → Save and new.
- Data Type : **Picklist**
Field Label : Role
Field Name : Auto Populates
Value : Enter value, with each value separated by a new line. Batsman
Bowler
Batsman & Wicket Keeper
All Rounder
- Data Type : **Picklist**
Field Label : Batting Style
Field Name : Auto Populates
Value : Enter value, with each value separated by a new line.
Right- Handed
Left - Handed
Click on Next → Next → Save and new.
- Data Type : **Picklist**
Field Label : Bowling Style
Field Name : Auto Populates
Value : Enter value, with each value separated by a new line.
Right-Arm Fast
Left-Arm Fast

Right-Arm Medium
Left-Arm Medium
Right-Arm Spin
Left-Arm Spin

Click on Next → Next → Save and new.

- Data Type : **Picklist**

Field Label : Nationality

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line.

Australia
Bangladesh England
India
New Zealand
Pakistan
South Africa Sri
Lanka
West Indies
Afghanistan
Zimbabwe
Ireland
Scotland
Netherlands
United Arab Emirates
Nepal
United States of America

Click on Next → Next → Save and new.

- Data Type : **Picklist**

Field Label : Team

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

Chennai Super Kings (CSK)
Delhi Capitals (DC)
Kolkata Knight Riders (KKR)
Mumbai Indians (MI)
Punjab Kings (PBKS)
Rajasthan Royals (RR)
Royal Challengers Bangalore (RCB)
Sunrisers Hyderabad (SRH)
Gujarat Titans (GT)
Lucknow SuperGiants (LSG)

- Data Type : **Formula**

Field Label : AVG Score

Field Name : Auto Populates

Formula Return : **Number**

Formula : IF(
Matches_Played__c > 0,
Total_Runs__c / Matches_Played__c ,
0
)

Simple Formula Advanced Formula

Insert Field Insert Operator

AVG Score (Number)=

```
IF(
    Matches_Played__c > 0,
    Total_Runs__c / Matches_Played__c ,
    0
)
```

Check Syntax No syntax errors in merge fields or functions. (Compiled size: 130 characters)

Click on Next → Next → Save & New.

- Data Type : **Lookup Relationship**

Related To : Match Performance

Field Label : Match Performance

Field Name : Auto Populates

Click on Next → Next → Save.

Activity 2 : Create Custom fields for Match Performance object.

1. Data Type : **Text**

Field Label: Match Performance Name

Field Name : Auto Populates

Length : 80

Click on Next → Next → Save and new.

2. Data Type : **Checkbox** Field Label: Player of the Match Field Name : Auto Populates

Click on Next → Next → Save and new.

3. Data Type : **Picklist**

Field Label: Match Result

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

WON

LOST

DRAW

Click on Next → Next → Save and new.

4. Data Type : **Number**

Field Label: Runs Scored

Field Name : Auto Populates

Click on Next → Next → Save and new.

5. Data Type : **Number**

Field Label: Balls Faced

Field Name : Auto Populates

Click on Next → Next → Save and new.

6. Data Type : **Number**

Field Label: Fours

Field Name : Auto Populates

Click on Next → Next → Save and new.

7. Data Type : **Number**

Field Label: Sixes

Field Name : Auto Populates

Click on Next → Next → Save and new.

8. Data Type : **Number**

Field Label: Overs Bowled

Field Name : Auto Populates

Click on Next → Next → Save and new.

9. Data Type : **Number**

Field Label: Wickets Taken

Field Name : Auto Populates

Click on Next → Next → Save and new.

10. Data Type : **Number**

Field Label: Runs Conceded

Field Name : Auto Populates

Click on Next → Next → Save and new.

11. Data Type : **Number**

Field Label: Maiden Overs

Field Name : Auto Populates

Click on Next → Next → Save and new.

12. Data Type : **Number**

Field Label: Stumpings

Field Name : Auto Populates

Click on Next → Next → Save and new

13. Data Type : **Number**

Field Label: Catches

Field Name : Auto Populates

Click on Next → Next → Save and new.

14. Data Type : **Master - Detail Relationship**

Related To : Player

Field Label : Player

Field Name : Auto Populates Click
on Next → Next → Save.

Milestone 6 : Roll-up Summary

Activity 1 : Create Roll-up Summary for Player object.

Note :- Roll-up Summary can only be created on parent object in Master- Detail Relationship.

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,

6. Choose a data type as **Roll-up Summary**. Click Next.

Field Label: Matches Played

Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : COUNT

Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field

Label: Number of 50's

Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : COUNT

Filter Criteria : Only records meeting certain criteria should be included in the calculation.

Click on Next → Next → Save and new.

The screenshot shows a configuration interface for a Roll-up Summary field. It is divided into two main sections: 'Select Roll-Up Type' and 'Filter Criteria'.

Select Roll-Up Type: This section has a light blue header. It contains four radio buttons: 'COUNT' (which is selected), 'SUM', 'MIN', and 'MAX'. To the right of these buttons is a dropdown menu labeled 'Field to Aggregate' with the value '--None--'.

Filter Criteria: This section has a light blue header. It contains two radio buttons: 'All records should be included in the calculation' (which is unselected) and 'Only records meeting certain criteria should be included in the calculation' (which is selected).

Below the selected radio button is a table for defining filter criteria:

Field	Operator	Value	
Runs Scored	greater or equal	50	AND
Runs Scored	less than	100	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

Below the table, there is a small note: 'For checkbox fields, enter a value of True for checked or False for not checked. For picklist fields, enter the master picklist field value in your corporate language.'

At the bottom of the section are two buttons: 'Save' and 'Cancel'.

- Data Type : Roll-up Summary Field

Label: Number of 100's

Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : COUNT

Filter Criteria : Only records meeting certain criteria should be included in the calculation.

Click on Next → Next → Save and new.

Master Object: Player
Summarized Object: Match Performances

Select Roll-Up Type

☒ COUNT
☐ SUM
☐ MIN
☐ MAX

Field to Aggregate: --None--

Filter Criteria

☐ All records should be included in the calculation
☒ Only records meeting certain criteria should be included in the calculation

Field	Operator	Value	
Runs Scored	greater or equal	100	AND
Runs Scored	less than	200	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		

For checkbox fields, enter a value of True for checked or False for not checked. For picklist fields, enter the master picklist field value in your corporate language.

Previous Next Cancel

Milestone 7 : Flows

Activity 1 : Screen Flow

1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.

Search Setup

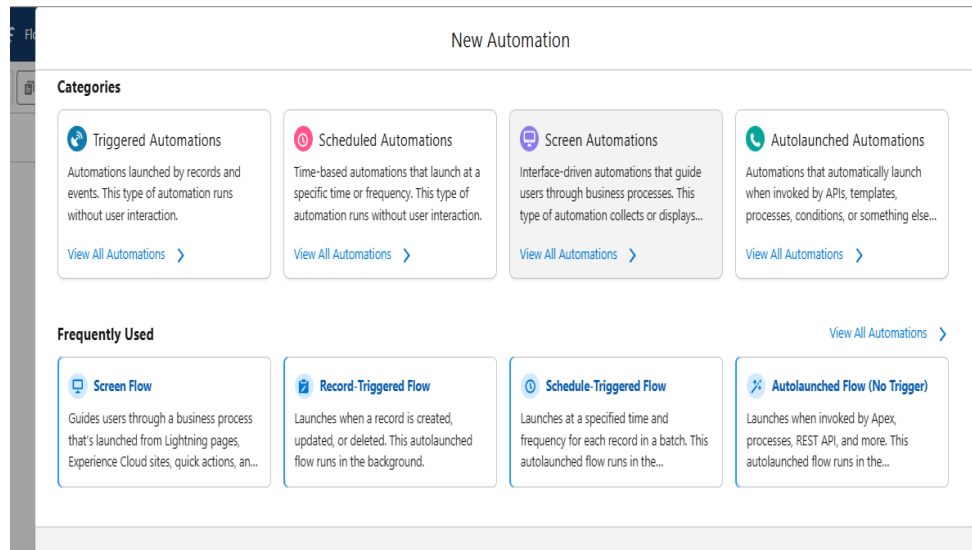
ager ▾

 **SETUP**
Flows

[Flow Trigger Explorer](#) [New Flow](#)

Try the [Automation Lightning App!](#)

2. Select Screen Flow and click Create.



3.

4. Click (+) Icon on the flow builder and Select **Screen Component**.

Label : Players screen

API : Players_screen

Select the **Picklist** component and place it on the screen.

Label : Team

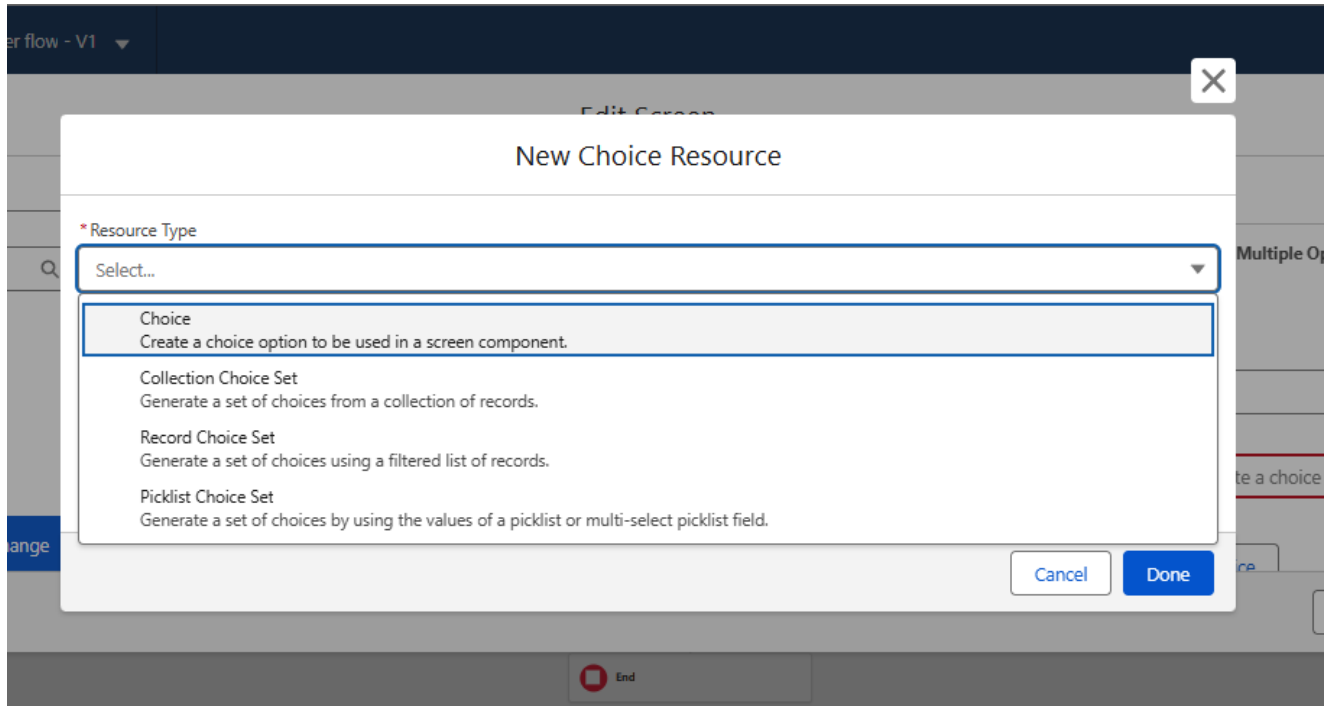
API : Team

From Component Pane on the left side select **Component Type** : Picklist

Data Type : Text

Choice : Click New Choice Resources

Resource Type : Picklist Choice Set



API Name : Team
Object : Player
Data Type : Picklist
Field : Team Click
 Done.

New Choice Resource

* Resource Type
 Picklist Choice Set

* API Name
 Team

Description

* Object
 Player

* Data Type
 Picklist

* Field
 Team__c

Sort Order
 Default Order of Field

Cancel Done

4. Click (+) Icon on the flow builder and Select **Get Records Component**.
Label : Players

API Name : Players

Object : Player

Filter player records :

Field : Team

Operator : Equals

Value : Click New Resource

Resource Type : Variable

API Name : Dynamic

Data Type : Text

Default Value : Screen component : Team (Picklist)

Get Records

Filter player Records

Condition Requirements

All Conditions Are Met (AND) ▼

Field	Operator	Value
Team X	Equals ▼	dynamicvalues X

+ Add Condition

Sort player Records

5. Click (+) Icon on the flow builder and Select **Screen Component**.

Label : Player_info

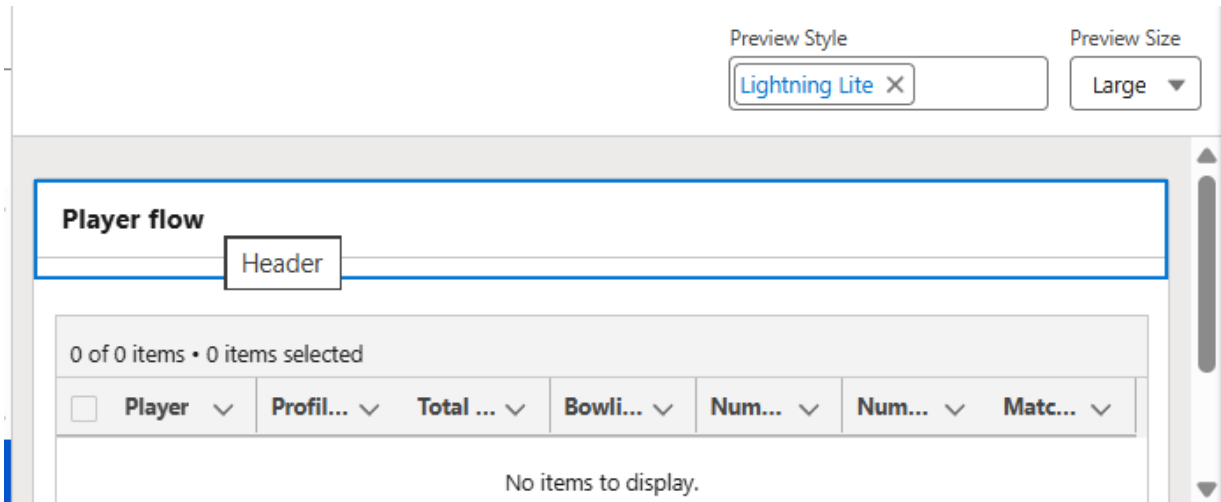
API Name : Player_info

From Component Pane on the left side select **Component Type :** Data table

API Name : Player_info_table

Label : Data Table

Configure Data Source : Source Collection : Players from players (Player__c)

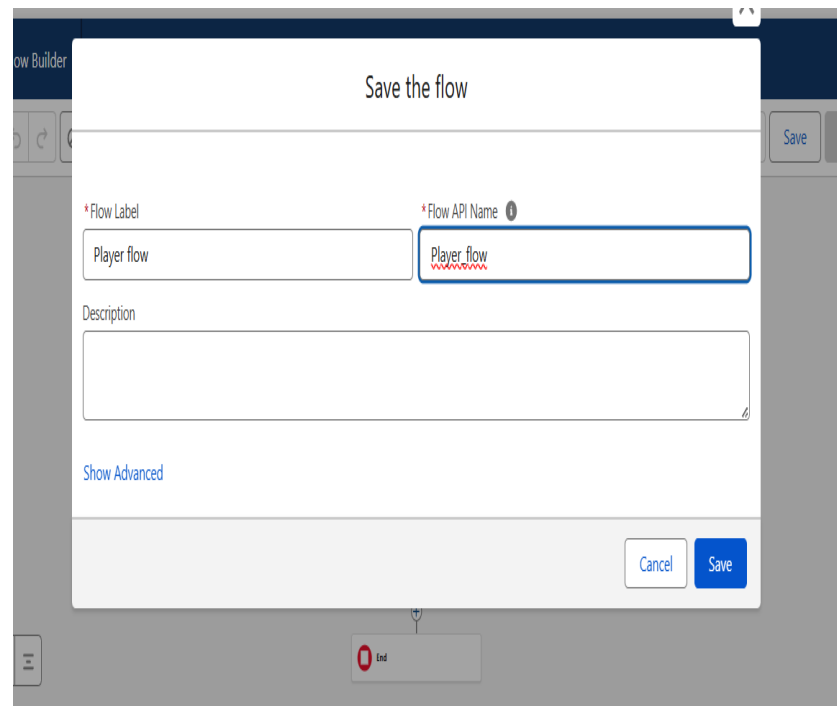


In Configure Columns : Add all the required Columns

Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style | Matches Played | Number of 50's | Number of 100's Click Done.
Click Save.

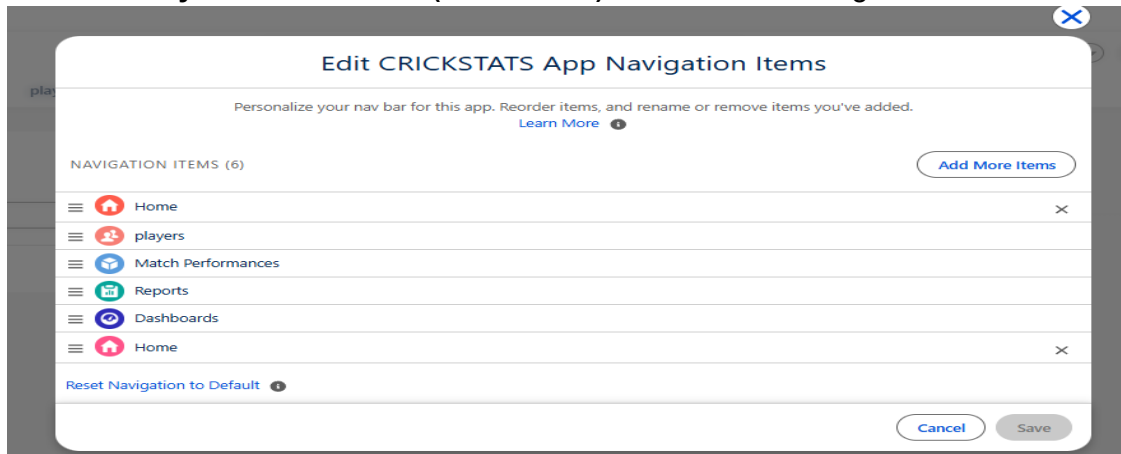
Flow Label : Player Flow

Flow API Name : Player_Flow

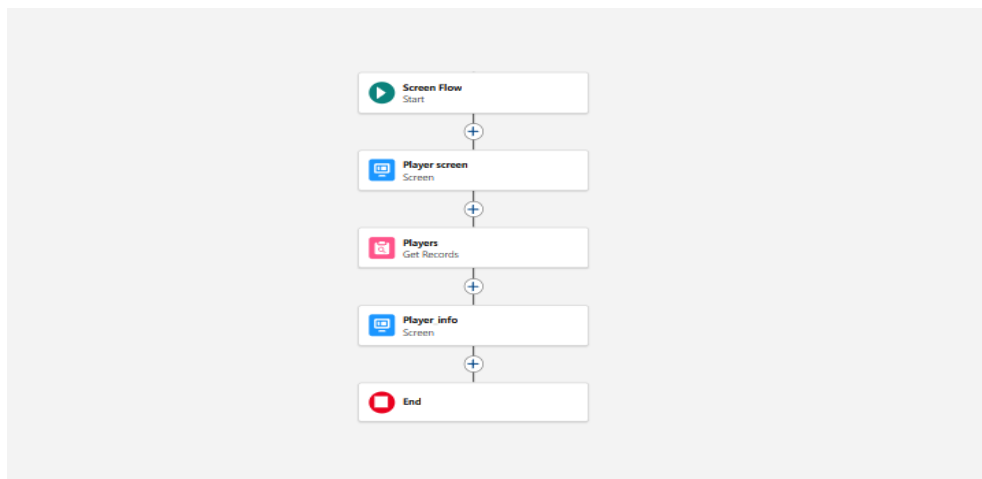


Activity 2 : Flow on Home Page

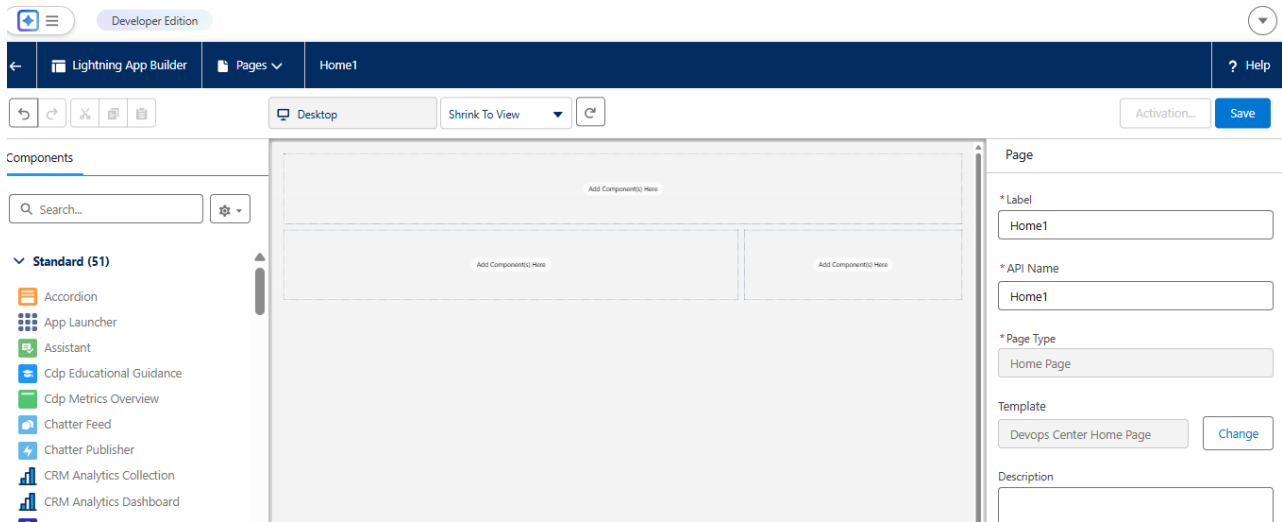
1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.



2. Remove all other components and Add **FLOW** component from the standard components on the leftside.



3. **Flow** : Players Flow



4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to App**.
6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.

FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized system
- Real-time analytics
- Reduced manual work

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion:

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

10. FUTURE SCOPE

- Player ranking system
- Public statistics portal
- Multi-league support

11. APPENDIX

- Salesforce Objects & Fields
- Manual match statistics
- Flow Configurations

Thank You

